Arjan Faber

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Passionate and detail-oriented Software Developer with experience in ROS/ROS2 with C/C++ and Python programming, Linux-based systems, and automation. Strong background in developing and testing software for real-time systems, with expertise in ROS, CUDA, and multi-modal sensor integration. Proven track record in automating test environments, debugging, and optimizing systems, with a deep interest in automotive technologies and embedded software.

EDUCATION

University of Edinburgh — **Edinburgh, United Kingdom**
Master of Science in Data Science with High Performance Computing — September 2023 - August 2025

- Dissertation at School of Informatics. Topic: *"Verifying the upper confidence bound algorithm (UCB) in Isabelle/HOL"*.
- Current GPA: Merit (61.5%), 40/90 ECTS.

Politecnico di Milano — **Milan, Italy**
Extracurricular graduate course in Mathematical Engineering — September 2024 - December 2024

- Course name: Computational Fluid Dynamics (worth 10 ECTS (not participated in the examination)).
- Curriculum: Numerical methods for fluid dynamics, Navier-Stokes equations and discretization techniques.

University of Groningen — **Groningen, Netherlands**
Bachelor of Science in Econometrics & Operations Research — January 2016 - July 2023

- Thesis topic : "*A comparison of machine learning methodologies for proxying CDS spreads*", awarded 8.0/10.
- Core statistics & econometrics courses GPA: 7.6/10 (70 ECTS). Overall GPA 6.7/10 (180 ECTS).
- Courses: Linear Algebra, Multivariate Calculus, Difference & Differential Equations, Programming in C++, Stochastic Models, Econometrics, Operations Research.

WORK EXPERIENCE

Human Media Interaction (HMI) Group, University of Twente — **Enschede, Netherlands**
Machine Learning Researcher (full-time contract) — January 2024 – July 2024

- Implemented deep Q-learning (DQN) and inverse reinforcement learning (MaxEnt IRL) to enable Kuka robots to acquire social behaviors in real-time, adapting to human feedback through multimodal cues.
- Led a data collection study using advanced sensors (LiDAR, RealSense cameras, Kinect) to gather over 25GB of interaction data. Implemented **OpenPose** and **CUDA** in **ROS Noetic (C++ & Python)** to process body and facial expressions for the robot's ability to interpret and respond to human behaviors.

Keywords: https://harmony-eu.org/, https://arjfaber.github.io/publications/

Achmea — **Apeldoorn, Netherlands**
Quantitative Modeller Intern — September 2021 – January 2022

- Converted modules for a stochastic model in MATLAB to Python code.
- Implemented a variant of the **Black-Scholes** model in Python for pricing call options.
- Attended daily meetings with the BI & Data team for automation of processes.

Keywords: Indexation, Options Pricing.

RESEARCH EXPERIENCE

School of Informatics, University of Edinburgh — **Edinburgh, United Kingdom**
Graduate researcher — February 2025 - Present

Dissertation topic: *"Verifying the upper confidence bound algorithm (UCB) in Isabelle/HOL"*. Under supervision of prof. dr. Wenda Li, School of Informatics. Key focus areas of my dissertation:

- Formalizes measure-theoretic probability in Isabelle/HOL to verify stochastic decision-making algorithms.
- Verifies concentration inequalities and tail bounds for UCB's regret guarantees.
- Extends martingale theory for regret analysis in multi-armed bandits.
- Formalizes the Optional Stopping Theorem and Martingale Central Limit Theorem in Isabelle.
- Represents continuous-time stochastic processes for continuous reward bandit problems.
- Expands formal verification to reinforcement learning models.

Department of Economics, Econometrics and Finance, University of Groningen — **Groningen, Netherlands**
Undergraduate thesis student — September 2020 – January 2021

Thesis topic: *"A comparison of machine learning methodologies for proxying CDS spreads"*. Achievements:

- Transformed a single hidden layer Artificial Neural Network (ANN) into a **Bayesian Neural Network (BNN)** by considering the idea of weights and biases uncertainty introduced by Blundell et al. (2015).
- Introduced **local interpretable model-agnostic explanations (LIME)** to improve interpretability of the methodologies.

Keywords: Tensorflow, ProbFlow, Scikit-Learn, LIME.

EXTRACURRICULAR ACTIVITIES

Econometric Study Association VESTING **Groningen, Netherlands**
Committee chairman & external affairs officer September 2017 – May 2019

- Co-responsible for organizing multiple sports tournaments and the participation of the study association in the world's largest student relay race.
- Co-responsible for the acquisition of foreign commercial partners for a study trip to Taipei City, Taiwan. Established partnership with Taiwan Stock Exchange, Deloitte, Willis Towers Watson and National Taiwan University.

Keywords: Team work, Communication skills, Cross-cultural.

SKILLS, ACTIVITIES & INTERESTS

Languages: Native in Dutch, fluent in English, basic German (reading).
IT Skills: Python, C, C++20 (memory management, Clang, Google Test), MATLAB, R, ReactJS, ROS1/ROS2, OpenMP, MPI, Linux, LaTeX, SLURM, ARCHER/ARCHER2 (HPC, job scheduling, resource management).
Interests: Jazz enthusiast (guitar), Formula 1, traveling (Asia, Scandinavia).